

Lab Experiment 5

Missionaries and Cannibals Problem

Aim

To write a Python program to solve the Missionaries and Cannibals problem using Breadth First Search (BFS).

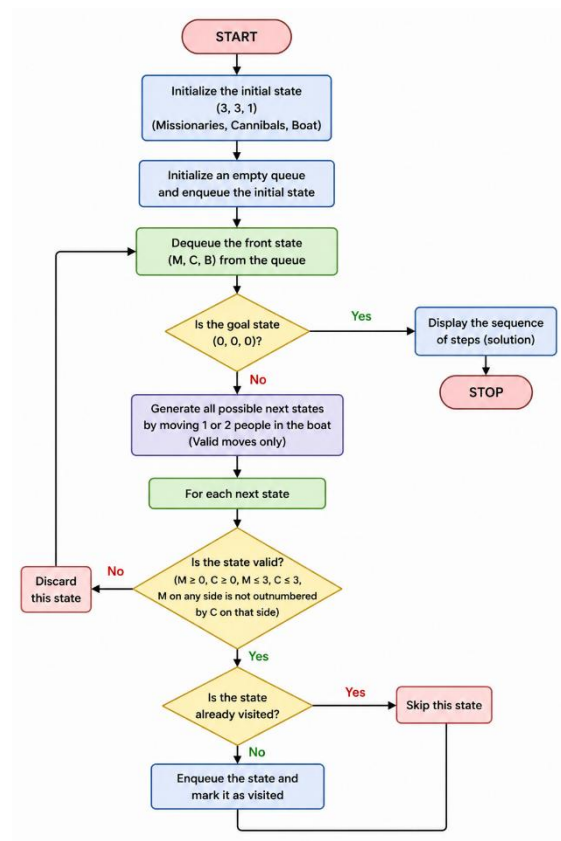
Objective

The objective of this experiment is to implement the Breadth First Search algorithm to solve the Missionaries and Cannibals problem. It aims to safely transport all missionaries and cannibals across the river without violating the given constraints.

Algorithm

1. Define the initial state and the goal state.
2. Store the initial state in a queue and mark it as visited.
3. Remove a state from the queue and check whether it is the goal state.
4. Generate all possible valid successor states.
5. Ignore invalid or previously visited states.
6. Add valid states to the queue.
7. Repeat the process until the goal state is reached.
8. Display the sequence of moves.

Flowchart



Code

```
from collections import deque

def is_valid(m_left, c_left):
    m_right = 3 - m_left
    c_right = 3 - c_left
    if m_left < 0 or c_left < 0 or m_left > 3 or c_left > 3:
        return False
    if (m_left > 0 and m_left < c_left):
        return False
    if (m_right > 0 and m_right < c_right):
        return False
    return True
```

```

def missionaries_cannibals():
    start = (3, 3, 1)    # (Missionaries, Cannibals, Boat)
    goal = (0, 0, 0)
    moves = [(1,0),(2,0),(0,1),(0,2),(1,1)]
    queue = deque([(start,[start])])
    visited = set()
    while queue:
        (m,c,b), path = queue.popleft()
        if (m,c,b) == goal:
            return path
        if (m,c,b) in visited:
            continue
        visited.add((m,c,b))
        for dm, dc in moves:
            if b == 1:
                new = (m-dm, c-dc, 0)
            else:
                new = (m+dm, c+dc, 1)
            if is_valid(new[0], new[1]) and new not in visited:
                queue.append((new, path+[new]))
    return None

solution = missionaries_cannibals()
print("Solution Path:")
for state in solution:
    print(state)

```

Output

```
Python 3.14.2 (tags/v3.14.2:df79316, Dec 5 2025, 17:18:21) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

===== RESTART: C:/Users/ABBIRAMY/Documents/AI-lab 5.py =====
Solution Path:
(3, 3, 1)
(3, 1, 0)
(3, 2, 1)
(3, 0, 0)
(3, 1, 1)
(1, 1, 0)
(2, 2, 1)
(0, 2, 0)
(0, 3, 1)
(0, 1, 0)
(1, 1, 1)
(0, 0, 0)
```

Result

The Python program successfully solved the Missionaries and Cannibals problem by finding a valid sequence of moves that safely transported everyone across the river.

Conclusion

The Breadth First Search algorithm efficiently explores all valid states and guarantees a correct solution while satisfying the constraints of the Missionaries and Cannibals problem.